Speech-to-Text Accuracy Report – Zecpath AI

Objective

To evaluate speech-to-text accuracy across different accents, noise levels, and speaking styles.

Test Dataset Summary

Test Type	Samples
Clean Audio	20
Noisy Background	20
Indian Accent	20
Mixed Accent	20
Fast Speech	10
Interrupted Speech	10
Total	100

Accuracy Results

Condition	Accuracy
Clean Audio	96%
Indian Accent	91%
Mixed Accent	88%
Noisy Background	82%
Fast Speech	85%
Interrupted Speech	80%

Overall Accuracy

Average Accuracy = 87%

Error Types

Error Type	Example
Misheard words	"node" → "note"
Missing punctuation	No sentence breaks
Filler noise	"um", "uh"
Broken sentences	Partial phrases

Improvements Applied

- Filler word removal
- Punctuation correction
- Silence detection
- Text normalization

Sample Input

um i have uh 3 years experience in python and django

Sample Output

I have 3 years experience in python and django.

Future Improvements

- Real-time streaming STT
- Accent adaptation models
- Context-aware punctuation